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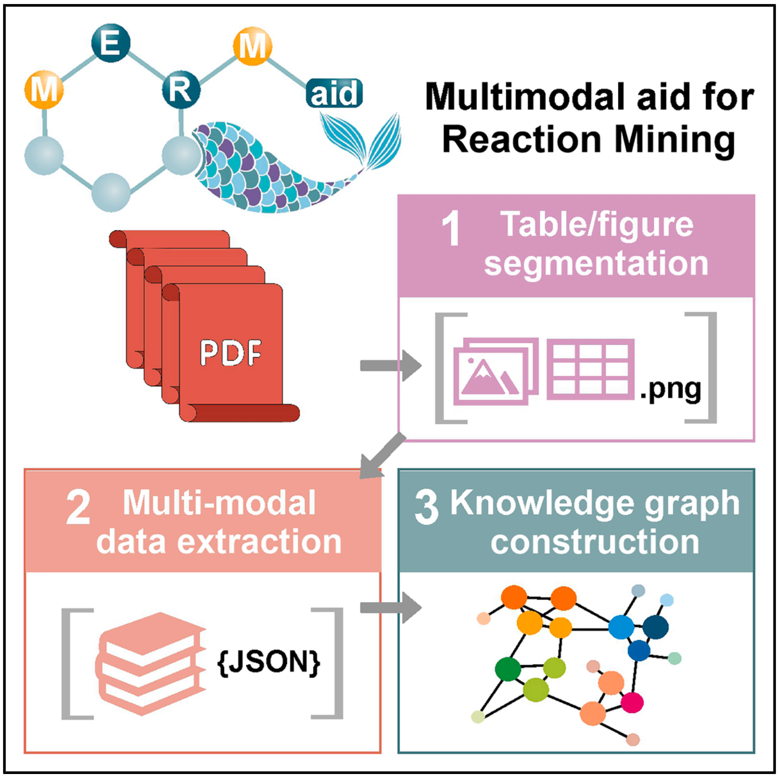
MERMaid: Universal multimodal mining of chemical reactions from PDFs using vision-language models

Graphical abstract



Highlights

- End-to-end pipeline that converts cross-PDF graphics into a knowledge graph
- Modular VLM-powered modules for image segmentation, parsing, and graph generation
- 87% end-to-end accuracy on 100 articles from three different chemical domains
- Robust to diverse layouts and stylistic variability

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In brief

MERMaid is a modular, vision-language model-powered pipeline for automated reaction mining from scientific PDFs. It segments and parses figures, tables, and captions to extract chemical data and reformulates it into machine-readable knowledge graphs. Demonstrated across organic electrosynthesis, organic synthesis, and photocatalysis with 87% accuracy, MERMaid enables structured data extraction from previously inaccessible visuals, supporting data-driven research, trend analysis, and integration with autonomous laboratory systems. It can be readily adapted for reaction mining across chemical disciplines.

Demonstrate

Proof-of-concept of performance with intended application/response

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CellPress

Article

MERMaid: Universal multimodal mining of chemical reactions from PDFs using vision-language models

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PROGRESS AND POTENTIAL Despite today's increasingly data-driven research paradigm, much of past foundational knowledge is locked in legacy formats such as PDFs. Automating data digitization of scientific literature remains challenging due to PDF variability, complex visual content, and the need to integrate multimodal information. MERMaid (multimodal aid for reaction mining) overcomes this by using the visual cognition and reasoning capabilities of multimodal AI models to transform graphical elements in PDFs into machine-actionable knowledge. We demonstrate MERMaid's practical utility and its topic-agnostic nature for multiple chemical domains. Its modular and composable architecture allows for independent deployment and seamless integration of additional capabilities. We believe our work will contribute significantly toward AI-powered scientific discovery, unlocking legacy knowledge for today's researchers and enabling integration with knowledge-intensive systems such as self-driving labs.

SUMMARY

Data digitization of scientific literature is essential for creating machine-actionable knowledge bases to advance data-driven research and integrate with self-driving laboratories. It is especially critical to extract, interpret, and structure data from graphical elements, the primary medium for conveying complex scientific insights. However, this remains challenging due to the inherent lack of semantic structure in the prevalent PDF format, the complexity of visual content, and the need for multimodal integration. We present MERMaid (multimodal aid for reaction mining), an end-to-end pipeline that converts disparate visual data across PDFs into a coherent knowledge graph. Leveraging the emergent visual cognition and reasoning capabilities of vision-language models, MERMaid demonstrates chemical context awareness, self-directed context completion, and robust coreference resolution to achieve 87% end-to-end accuracy across three chemical domains. Its modular design facilitates future application to diverse data beyond reaction mining, promising to unlock the full potential of scientific literature for knowledge-intensive applications.

INTRODUCTION

Leveraging prior knowledge is crucial for accelerating AI-driven scientific understanding and discovery. Specific applications include enabling data-driven research, eliminating cold-start problems with expert knowledge bases in autonomous systems such as self-driving laboratories, and training data-hungry AI models effectively.[1–5] However, the main bottleneck is the accessibility of high-quality, structured, and machine-actionable scientific data in an ingestible form for AI models and chemical databases. Significant efforts have been made to construct databases, with pioneering examples being the Harvard Clean Energy Project,[6] the AFLOW database,[7] the Materials Project,[8] and the Open Quantum Materials Database.[9] More recent advances have introduced additional theoretical databases such as the Open Materials 2024 Inorganic Materials Dataset[10] and the Quantum MOF database,[11] as well as experimental databases such as the Open Reaction Database (ORD).[12] Despite these efforts, a wealth of chemical and materials information remains locked in legacy journal articles and patents.

To harness these previously inaccessible data, chemical data digitization, i.e., the conversion of chemical information into machine-readable formats, is essential. This is particularly true for graphical elements such as figures, tables, and diagrams, which are a primary medium for conveying dense, high-value scientific content. However, digitizing data from these visual elements remains a major challenge due to three main hurdles. First, accurate extraction requires joint segmentation of graphical elements and their associated captions or headings, as key contextual information is often split between the two. This is especially difficult in PDFs, the dominant format for scientific publishing, because they lack the structural features and semantic metadata tags that typically facilitate automated parsing.[13] They also exhibit high variability in formatting. These issues are amplified in supplemental materials, which often contain critical experimental details but follow no standardized format. Second, processing scientific graphical elements demands multimodal reasoning and domain-specific contextual understanding. They often contain symbolic representations, such as circuit symbols or physics notations, that require scientific interpretation and cannot be resolved through basic visual analysis alone.[14,15] Additionally, scientific tables often present multiple experimental entries with partial information, requiring the ability to infer and align missing context across related entries. This need for self-directed context completion within and across graphical elements remains challenging for current models. Third, structuring and standardizing the extracted content into machine-actionable big data formats is essential to ensure compatibility with downstream analyses.[16–18] Without a robust automation pipeline, raw extracted data often require extensive manual cleaning and expert-driven postprocessing. Effective data organization must not only enforce standardization but also resolve ambiguities, normalize terminologies, and integrate disparate data into a coherent framework. Existing tools such as ChemDataExtractor,[19] ChemDataExtractor 2.0,[20] and PDFDataExtractor[21] focus only on text and lack true multimodal capabilities or adaptability across different extraction contexts. More recently, OpenChemIE[22] introduced multimodal parsing but relies on LayoutParser,[23] which is not optimized for scientific documents, leading to poor information capture. Like its predecessors, OpenChemIE also requires significant model retraining to adapt to new extraction tasks, limiting scalability and extensibility.

This work presents MERMaid (multimodal aid for reaction mining), our universal, end-to-end chemical data digitization platform that identifies, segments, and converts figures, schemes, and tables from scientific literature into structured, machine-readable reaction schemas for seamless integration with various database formats such as knowledge graphs (Figure 1). Building on insights from our prior work in multimodal parsing,[15] we have developed additional custom modules to enhance automated knowledge retrieval and organization.

(1) VisualHeist is a fine-tuned toolkit built on the top of Microsoft's Florence-2 vision foundation model[24] for joint segmentation of figures, schemes, and tables from PDFs, along with their accompanying captions, headings, and footnotes. It demonstrates robust performance (≥93%) across a wide range of formatting styles and publication time periods.
(2) DataRaider is a vision-language model (VLM)-powered tool designed to extract specific information associated with user-defined keys. By leveraging OpenAI's GPT-4o VLM,[25] it converts the extracted images into structured reaction dictionaries. Notably, it is capable of footnote-label resolution and successfully associates reference labels within figure images with their textual footnote descriptions. It also supports self-directed context completion by aligning different reaction entries presented in tables even when not all reaction conditions are reported for each entry, allowing the pipeline to handle incomplete or context-dependent information, a common challenge in scientific literature.
(3) KGWizard functions as a text-to-database engine, reformulating the reaction dictionaries into knowledge graphs. The conversion to knowledge graphs offers the advantage of synthesizing knowledge across large datasets, facilitating the downstream discovery of new relational insights between reactions rather than treating each as an isolated instance. To standardize different abbreviations and common names used across different literature to a single entity and avoid duplicate entities, we employ a two-pronged approach utilizing the PubChem database and retrieval-augmented generation (RAG)[26–28] for explicit and implicit coreference resolution, respectively.

MERMaid addresses key existing limitations to yield complete reaction datasets with >92% accuracy, and subsequent conversion to an interconnected knowledge graph with 96% accuracy, maintaining excellent data integrity. Our pipeline is topic-agnostic and adaptable to various chemical domains. Here, we demonstrate its application in extracting reaction data for organic electrosynthesis, organic synthesis, and photochemical reactions, each characterized by domain-specific terminology. MERMaid is modular and extensible, whereby individual modules can be deployed independently for their specific functions, and new modules, such as text processing capabilities and optical chemical structure recognition (OCSR) tools, can be added to expand the framework to data beyond reaction data mining or support alternative formats such as the ORD. Complementary to prior large language model (LLM)-based efforts that focus primarily on text mining,[29–32] MERMaid tackles two foundational bottlenecks: the inaccessibility of visual data and the fragmentation

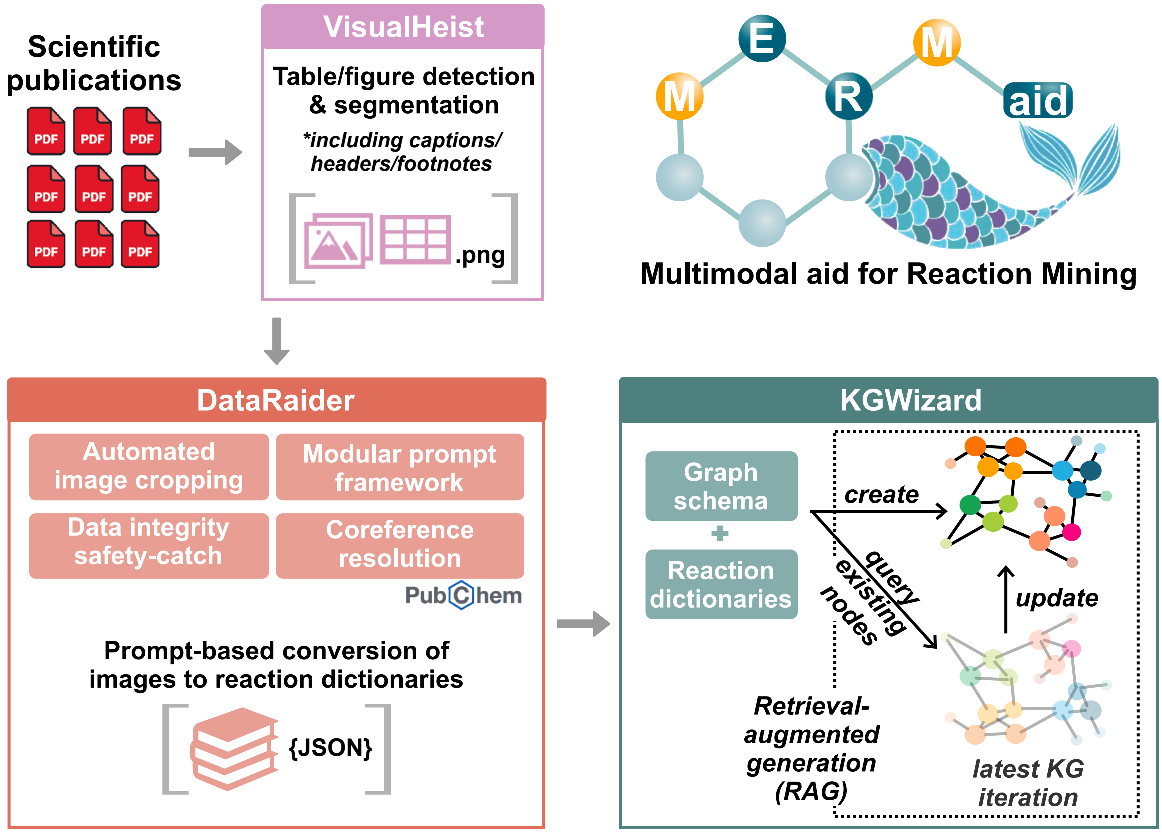

Figure 1. Schematic overview of MERMaid, which integrates three sequential modules for joint figure-caption segmentation from PDFs, image-to-data multimodal analysis, and knowledge graph construction, respectively

of experimental information across documents. It unifies multimodal segmentation, VLM-based reasoning for image-based data, and structured knowledge integration into a coherent, scalable, and domain-agnostic pipeline. This positions MERMaid as foundational infrastructure to make chemical literature more interoperable with AI-guided platforms for data-driven scientific discovery. By enabling the reuse of its modular components and facilitating seamless expansion, we envision that MERMaid serves as a powerful pipeline to accelerate the digitization and integration of existing chemical knowledge for knowledge-intensive autonomous systems.

RESULTS

MERMaid features three sequential modules to perform the following: table and figure segmentation from PDFs, multimodal analysis to extract relevant information as structured reaction schema, and knowledge graph construction. Each module can be implemented as standalone tools, or integrated into automated knowledge ingestion pipelines, as demonstrated in this work.

VisualHeist for table and figure segmentation from PDFs

VisualHeist is a deep-learning-based model to segment figures, schemes, and tables from PDF documents, along with their captions, headings, and footnotes (Figure 2A). Inspired by the TF-ID model,[33] VisualHeist is fine-tuned from Microsoft's Florence-2 vision foundation model.[24] Our model is fine-tuned on a diverse dataset of 3,435 figures and 1,716 tables annotated from a variety of documents to ensure robust performance across different presentation styles. These include main publications, supplemental materials, and unformatted archive papers (full DOI list in Table S1).

When evaluated on a dataset of 121 PDFs from 5 publishers (full DOI list in Table S2), covering diverse subject areas ranging from organic and inorganic chemistry, atmospheric science, batteries, materials science, metal-organic frameworks (MOFs), biology, to science education, VisualHeist achieves excellent performance, with overall precision, recall, F1, and accuracy exceeding 93% across a total of 1,935 images (Figure 2B; Table S3). The performance metrics are calculated based on the following: true positive for correct segmentation, false positive for redundant segmentation (e.g., capturing journal logos or text information), and false negative for partial segmentation (e.g., missing portions of graphical images or captions). Notably, VisualHeist demonstrates robustness against highly unstructured supplemental materials, scoring ≥92% across all performance metrics on all 46 supplemental materials, even in more complex cases involving misaligned figure-caption pairs, high-density number of footnote descriptions, and large sentence spacings (Figures 2B and S1). It also scored ≥93% in all performance metrics on the 29 PDFs published before 2000, dating as

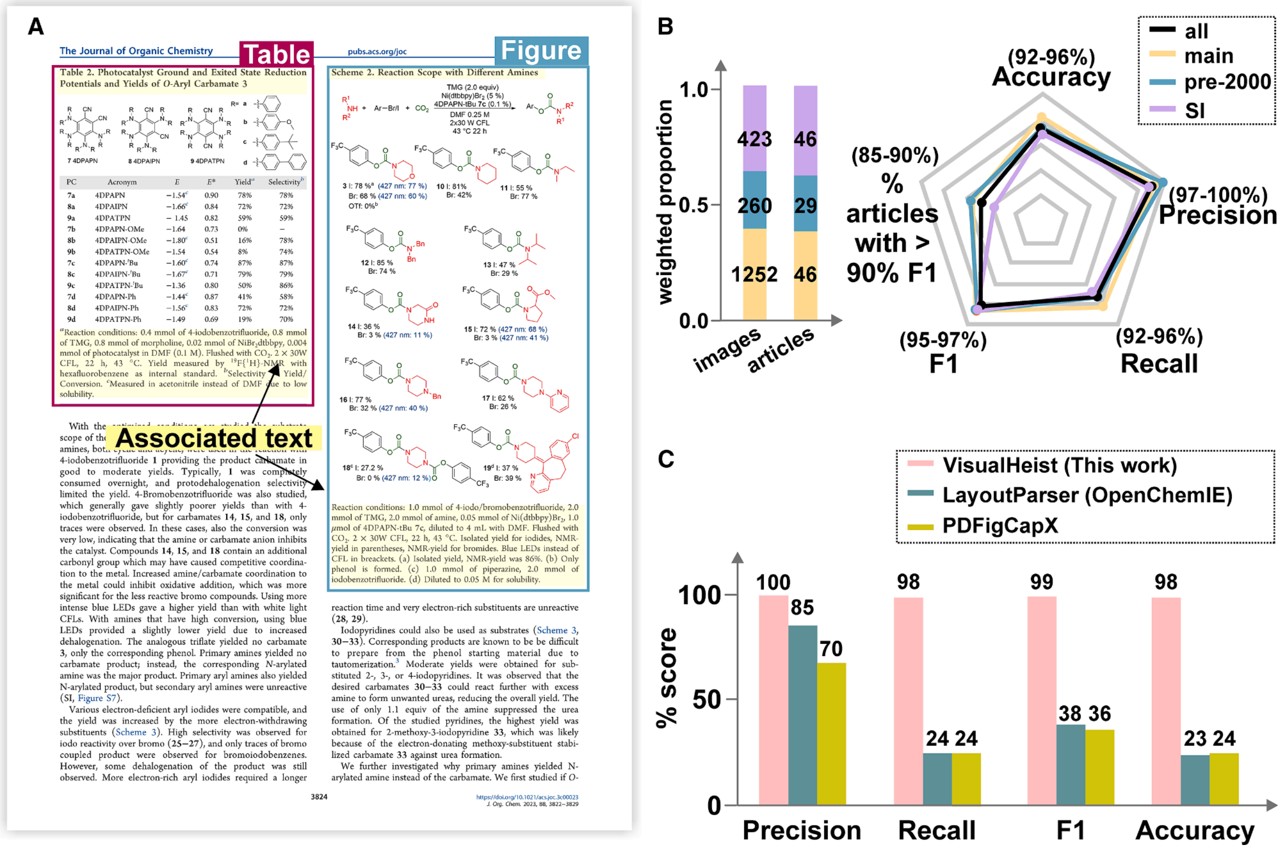

Figure 2. Performance evaluation of VisualHeist

(A) Example depiction of identification and segmentation of figures, schemes, and tables with their corresponding captions and headings from PDF documents. The example page is adapted with permission from Sahari et al.[34] Copyright 2023, American Chemical Society.

(B) Statistics on the number and distribution of images and PDF articles across different sources within the evaluation dataset (left), and evaluation of VisualHeist using various performance metrics (right).

(C) Performance comparisons of VisualHeist, LayoutParser and PDFigCapX using MERMaid-100 evaluation dataset.

far back as 1949 (Figure 2B). Missed cases primarily result from unconventional formatting choices in edge cases. For example, excessively tight spacing between consecutive images causes the heading of a later image being incorrectly captured as the caption for the previous image (Figure S1). Nonetheless, the model exhibits excellent performance in accurate image segmentation from varied data sources and time periods. Additionally, we investigated the model's robustness to resolution variability. In a resolution study where images were synthetically blurred at four increasing levels, VisualHeist maintains consistent accuracy (~96%), except for the highest level (Note S1).

We further compared the performance of VisualHeist against two existing PDF parser tools that provide comparable functionalities, namely OpenChemIE[22] and PDFigCapX,[35] both of which are designed for scientific documents. The evaluation was conducted using our curated benchmark dataset, MERMaid-100, which comprises 100 PDF articles across three distinct chemical domains: organic electrosynthesis, photocatalysis, and organic synthesis (full DOI list in Table S4). For OpenChemIE, it utilizes Layout Parser[23] for object detection, and captions are extracted as text using easyOCR.[36] To enable a fair comparison, we calculated the Jaccard similarity between the extracted and ground truth captions, which were manually compiled. Captions were considered fully captured if their Jaccard similarity exceeded a threshold of 0.70. Extracted captions that exceeded the ground truth length by more than 50 words were classified as false positives. On the other hand, PDFigCapX is designed to extract figures and captions from scientific biomedical documents. Each segmented graphical object and its corresponding textual description were evaluated jointly for both parsing tools, using the same performance metrics as described previously. VisualHeist outperformed OpenChemIE by 15%–75% and PDFigCapX by 28%–75% across all metrics (Figure 2C). Additional error analysis reveals that OpenChemIE demonstrates poor recall of only 24% due to under-segmentation and its inability to identify coreferences between segmented graphical objects and the relevant textual descriptions (Figure S2). As for PDFigCapX, its low recall primarily arises from missing figures, missing captions, and over-segmentation, where a single bounding box spans an

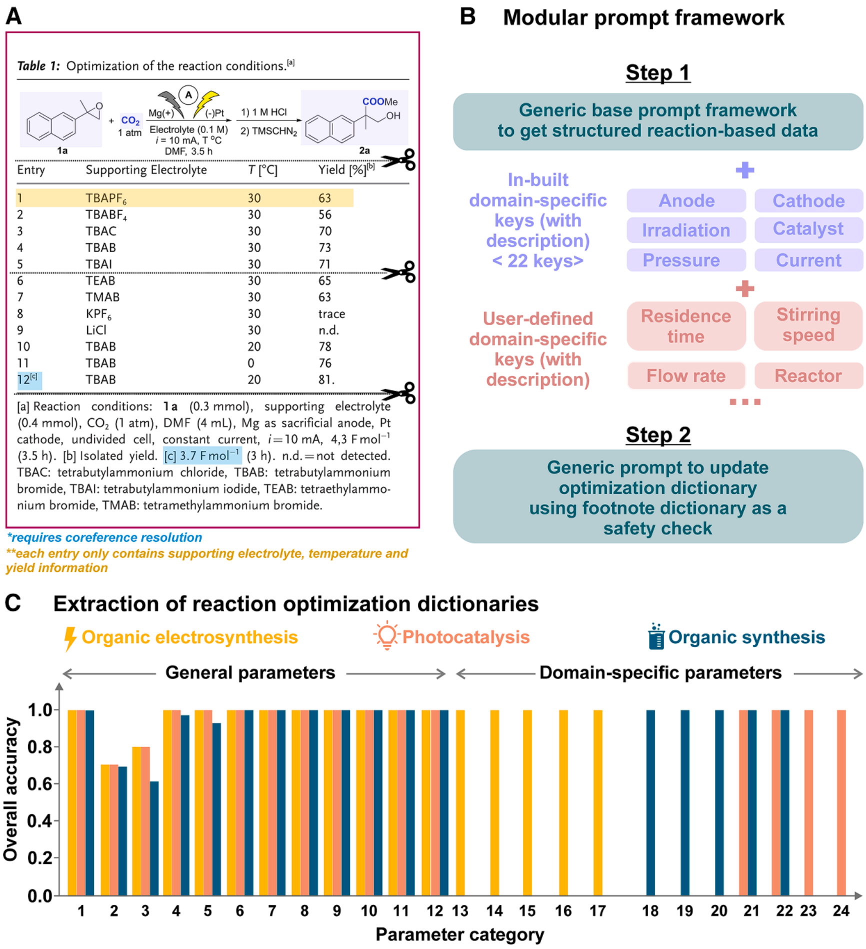

Figure 3. Performance evaluation of DataRaider

(A) Overview of the approach.

Optical chemical structure recognition of reactants and products is performed using RxnScribe (highlighted in purple); all other parameters are VLM-extracted. Images are cropped before analysis. Key challenges in organizing extracted data into meaningful output are highlighted. The figure used for illustrative purposes is adapted with permission from Zhang et al.[37] Copyright 2022, Wiley-VCH.

(B) Schematic of our modular, two-step prompt framework.

(C) Percent hard match identification accuracy for each parameter across three different chemical domains. Categories 1–24 refer to reactant reference indexes and quantities, reactant SMILES, product SMILES, solvent, solvent amounts, duration, air/inert atmosphere, temperature, yield, yield types, others, footnote references, anode, cathode, electrolyte, electrolyte amounts, current, catalyst, ligand, R-group substitutions, additives, additive amounts, irradiation conditions, and photocatalyst, respectively.

entire page and encompasses multiple images along with unrelated paragraphs (Figure S2).

DataRaider for image-based data mining

DataRaider takes in the segmented figure images as input and constructs structured reaction dictionaries from reaction diagrams and optimization tables through prompt engineering and automated image cropping (refer to VLM model selection). Based on our previous work, we have found that cropping figures into smaller subfigures before passing them collectively into the VLM mitigates information overload and ensures that no details are overlooked.[15] We thus adopt a similar strategy in this module (Figure 3A).

To minimize error propagation from earlier segmentation steps, DataRaider incorporates an internal validation mechanism that prompts the VLM to assess the quality of each segmented image. This check evaluates conditions such as missing or incomplete captions/headings or the presence of multiple conflicting labels. Only images that pass this validation step are processed further; flagged cases are excluded and logged with detailed error messages for user traceability. This lightweight filtering step enhances the robustness of the pipeline by ensuring that only high-quality inputs proceed to structured data extraction, achieving ≥97% precision, recall, F1, and accuracy on the same dataset used to evaluate VisualHeist (Table S5; full DOI list in Table S2).

A key feature of DataRaider is its modular, two-step, natural language prompt framework, which is designed to extract reaction optimization data across different fields (Figure 3B). In the first step, we construct a generic base prompt that specifies the expected data format for structuring the extracted reaction conditions as key-value pairs. This prompt also provides concise contextual information to: (1) prepare the VLM for coreference resolution, ensuring that reference labels in figures and tables are correctly associated with their corresponding textual descriptions in captions and headings and (2) enable autonomous "fill-in-the-blank" inferencing, where missing details are inferred from standard reaction conditions unless explicitly specified otherwise. This capability is especially crucial for parsing reaction optimization results, because reaction conditions are often only partially reported, typically as modifications to a set of standard conditions defined earlier. Most existing document processing toolkits struggle with this level of inference. The generic base prompt is complemented by 22 in-built keys defining different reaction parameters. These include common reaction categories such as solvent, duration, temperature, and atmosphere, as well as domain-specific keys such as anode, cathode, photocatalyst, and electrolytes (Table S6). Additionally, an “others” key is provided for any reaction parameter that does not fit into the predefined categories, allowing for more flexibility with complex reactions. Users can select only the relevant parameters in a straightforward, “plug-and-play” manner, and they can also add additional custom keys (tutorial provided in Note S2). The output from the first step is presented as two structured dictionaries: (1) a reaction dictionary, which contains the reaction conditions for each optimization run, and (2) a footnote dictionary, which stores the footnote identifiers and their corresponding descriptions (Note S3). In the second step, the formal prompt instructs the VLM to update the extracted reaction dictionary with information from the footnotes dictionary. This step serves as a safety check to ensure that any entry-specific changes specified in the footnotes, which may not have been incorporated in the first step, are correctly applied (Figure 3B). Finally, it is important to note that, in the current implementation, the reactants and products are retrieved as SMILES strings using RxnScribe,[38] a dedicated OCSR toolkit; while all other reaction parameters are extracted using GPT-4o, following our previously released MERMES framework.[15]

We evaluated the performance of DataRaider in constructing complete reaction optimization datasets using our MERMaid-100 dataset. The journal articles and their supplemental materials were downloaded in PDF format and contain a total of 104 image-captions and table-heading pairs relevant to our task of reaction optimization. Each model prediction was assigned as follows: (1) true positive for correctly identifying and classifying reaction parameters, (2) false positive for redundant information, (3) true negative for correctly identifying when reaction parameters were not described, and (4) false negative for missing information. It is important to note that we do not accommodate partial answers in cases involving mixed systems, such as reactions with mixed solvents or entries with multiple footnote references. A prediction is only considered a true positive when all associated footnote references and/or reported compounds are correctly identified. We also apply a strict hard-match criterion, meaning that the identified reaction parameter must be correctly matched to its intended role. For instance, an anode material annotated as the cathode is considered false negative. For the 12 general parameters, namely the reactant reference indexes and quantities, reactants, products, solvents, solvent amounts, duration, air/inert atmosphere, temperature, yield, yield type, others, and footnote reference labels, DataRaider achieves hard-match accuracies of >70% across all three chemical domains (Figure 3C; Table S7). Notably, the VLM-based extraction of reaction parameters (i.e., all parameters except reactant and product SMILES) reports high accuracies of >92% (Figure 3C; Table S7), with no observed performance degradation between figures extracted from main text and supplemental materials (Table S8), and across different modes of condition presentation (Note S4). For domain-specific reaction parameters, DataRaider similarly performs well, with accuracies reaching near-unity (Figure 3C; Table S7). These domain-specific reaction parameters include: anode, cathode, electrolyte, electrolyte amounts, and current for organic electrosynthesis (a total of five); catalyst, ligands, R-group substitutions, additives, and additive amounts for organic synthesis (a total of five); irradiation conditions, photocatalyst, additive, and additive amounts for photocatalysis (a total of four). Importantly, the module is robust in footnote-label coreference resolution and can accurately modify the correct reaction parameter value for a given entry or reaction run based on the associated footnote descriptions. Even in cases where footnote reference labels are correctly identified but the corresponding reaction parameter values are not updated after the first step, the second step ensures that these values are properly updated (Note S5). While existing reaction data extraction modules such as MolCoref in OpenChemIE and ChemDataExtractor 2.0 attempt to address coreference resolution, the sequence-generation and rule-based approaches employed by these pipelines are limited to resolving molecule-to-label coreferences and thus lack broader applicability. Lastly, the lower accuracies observed for reactant and product SMILES stem from inherent limitations in current OCSR tools (Table S9), not from the MERMaid pipeline architecture itself. Further discussion of this limitation is provided in limitations and future plans. It is also of note that DataRaider showed greater sensitivity to resolution variability compared with VisualHeist due to its reliance on fine-grained visual cues for accurate data extraction (Note S1).

KGWizard for knowledge graph construction

The final module in our knowledge ingestion pipeline, KGWizard, integrates the isolated reaction dictionaries into a highly interconnected database. This step is critical as it enables future downstream AI-driven exploration and analysis by: (1) facilitating efficient query mechanisms for data extraction and (2) supporting RAG models to enhance interactions with LLMs. Given the dynamic nature of chemical research and the efficiency of knowledge graph databases in data retrieval,[39–42] we chose a graph database for greater flexibility and scalability.

KGWizard leverages LLMs as a higher-order function to automate data type conversion. Given domain-specific instructions and transformation rules, the LLM dynamically synthesizes a parser that converts the raw input structure (e.g., a JSON dictionary) into a target data format (i.e., a graph database), which is represented as Python code (full implementation details in the methods). This enables the automatic construction of a tailored transformation workflow, ensuring that reaction entities, relationships, and metadata are accurately mapped while preserving domain semantics (Figure 4A). This approach significantly reduces the complexity and time required for database construction, and enables flexible, schema-adaptive conversion across different knowledge representation paradigms.

We began by defining a reaction graph schema (Y, Note S6). Using the extracted organic electrosynthesis data as an example, each reaction node in the database represents a single reaction run described in the original article (Figure 4D). These reaction nodes store key parameters of the reaction as property nodes (Figure 4B), contextualized by their edge notations (Figure 4C; Table S10). All reaction nodes associated with the same article are linked to a shared study node, creating a cohesive structure. This reaction graph schema is designed to be flexible and scalable, allowing for the addition of new property nodes or connections to refine and expand the contextual information for each reaction based on the data extracted from the article. It can also be easily adapted to other domains through minor modifications to the property nodes (Notes S2 and S7), enabling other scientists to extend its usability across a wider range of chemical fields. The raw reaction dictionaries can then be readily transcribed into the required graph database structure using the predefined schema template via an LLM.

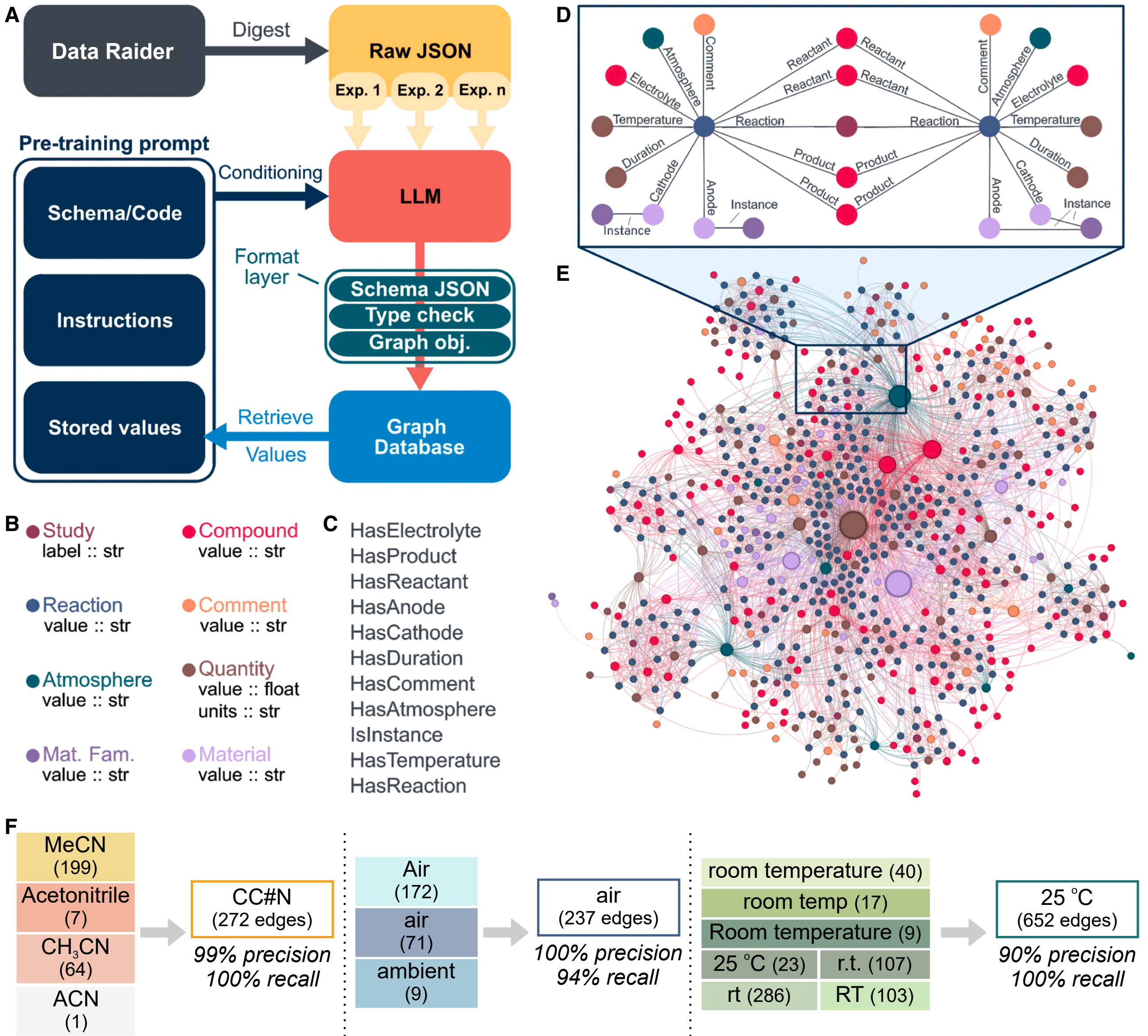

Figure 4. Key features of KGWizard

(A) KGWizard workflow.

(B) Node types of the graph database.

Attributes and their types after "::"; the attributes of red bold nodes are retrieved from the database to generate the conversion prompt.

(C) Edge types of the graph database.

(D and E) (D) Example of a reaction module, combining the environmental properties (nodes) and context (edges) from the (E) knowledge graph created from the organic electrosynthesis reactions within the MERMaid-100 dataset.

(F) Examples of coreference resolution.

A major challenge for chemical reaction data extraction is ensuring database consistency, as assimilating reaction information across different articles often introduces discrepancies and redundancies. The same compound may appear under different names or formats. For instance, acetonitrile can be labeled as "MeCN," "ACN," or "CH_3CN." This issue is further amplified in mixed-solvent systems, where authors use inconsistent delimiters, such as H_2O-MeOH, H_2O/MeOH, or H_2O:MeOH to describe the same water-methanol system. To address this issue, we employ a two-pronged approach for explicit and implicit coreference resolution. The former attempts to standardize all compound names to their canonical SMILES by querying the PubChem database via its PUG REST service.[43,44] The latter is done via a RAG approach that continuously queries the graph database for existing nodes with matching types holding values of the same context before creating new nodes. In doing so, we minimize duplicate nodes and enhance standardization. Additionally, the retrieval technique enables robust error handling for quality control; if the LLM generates a malformed output, the system can retrieve a correctly formatted version from the database and use it to refine the prompt.

For breadth, we tested KGWizard to automatically generate domain-specific knowledge graphs for each chemical domain in our MERMaid-100 dataset. Articles were processed in dynamic size batches (refer to methods), allowing the RAG module to incorporate preexisting data iteratively during each batch digestion and gradually update the database. Unique values from the generated nodes were included in subsequent prompts, enabling the model to verify whether an environmental variable already existed within the network. The resulting graphs comprise 1,470 nodes and 6,765 edges for the organic electrosynthesis dataset, 907 nodes and 4,439 edges for the organic synthesis dataset, and 846 nodes and 3,872 edges for the photocatalysis dataset (Figure 4E for organic electrosynthesis dataset; Figures S3 and S4 for the other two domains). Inspection of the knowledge graphs indicates a well-structured data architecture, with robust coreference resolution in unifying disparate entries with varying names for the same context into a singular property node. Notably, we achieved 96% averaged accuracy across nine frequently occurring values (Figure 4F; Table S11).

DISCUSSION

Limitations and future plans

To further improve accuracy, OCSR within the DataRaider module should extend beyond non-Markush and basic Markush structures, as graphical representations often contain more complex molecular structures. This remains an active area of research[45] and we are actively developing more robust OCSR capabilities as part of our roadmap to enhance MERMaid's ability.

In parallel, extending MERMaid to handle multi-step synthetic routes is another key milestone in our development roadmap. This will require more sophisticated segmentation strategies to accurately parse densely structured total synthesis schemes while maintaining the continuity of stepwise reaction context, as well as enhanced OCSR to resolve intermediates and structurally complex molecules.

The present implementation of KGWizard has three main limitations. (1) The graph schema is static; any modification requires a full rebuild of the knowledge graph, preventing incremental updates. (2) RAG is employed only for entity deduplication; extending it to supply richer context and improve translation fidelity lies outside the scope of this study. (3) As a direct consequence of (1) and (2), the schema cannot evolve automatically when new literature is ingested. Although recent work shows that LLMs can derive graph ontologies directly from raw data[46] and could support a fully dynamic, RAG-aware schema, we chose a user-defined approach to keep tight control over structure. This decision lets researchers organize information to match their experimental needs, ensuring coherence and facilitating downstream analysis. Developing an adaptive, LLM-driven workflow that updates the schema on the fly is therefore an important direction for future work.

It is also of note that using commercial models in this work accelerated development and reduced resource demands (with the exception of VisualHeist, which required training on a local workstation). However, fine-tuning open-source models such as LLAMA[47] and DeepSeek[48,49] offer promising alternatives for several steps in our workflow.[50,51] In line with our commitment to open science, we encourage other researchers to further expand our code, integrating custom models as needed.

Conclusions

A key challenge hindering the digitization of scientific literature is the extraction, interpretation, and structuring of data from information-dense graphical elements across disparate PDFs. MERMaid demonstrates the capability to transform image-based data from diverse chemical domains into machine-actionable formats with high efficiency and accuracy. Through strategic use of the cognition abilities of VLMs, image cropping, prompt engineering, and RAG techniques, we successfully enabled automated mining of chemical reactions to capture both the optimized reaction conditions and the associated optimization runs and studies with high fidelity, thereby providing a robust pipeline for augmenting chemical databases. Each module within MERMaid achieved robust individual performances, with averaged accuracies of 98%, 92%, and 96%, for VisualHeist, DataRaider, and KGWizard respectively, on our MERMaid-100 evaluation set. This level of accuracy surpasses the ~80% benchmark typically achieved by manual data extraction,[52] supporting its practical reliability for use in machine learning applications and integration into open-access chemical databases.

Although certain functionalities, such as OCSR, remain active areas for future improvement, MERMaid already provides a robust, general-purpose baseline for vision-based scientific data extraction, complementing current state-of-the-art systems that focus primarily on text-based extraction or domain-specific tasks. By addressing the largely untapped modality of visual data in scientific literature, MERMaid enables the automated extraction of experimental data from legacy documents that would otherwise remain inaccessible. Looking ahead, MERMaid contributes to our long-term goal of enabling data-driven discovery of novel scientific insights by tackling a foundational challenge: the pervasive data silos in scientific literature, where knowledge is fragmented across individual publications, limiting both human and machine reasoning. MERMaid addresses this challenge through robust multimodal information extraction and the integration of disparate data points into a consistent, structured knowledge framework. As an open-source platform, it further encourages community-driven extension and refinement, democratizing access to historical chemical data and supporting informed, data-centric decision making, and hypothesis generation. We anticipate that this work will contribute significantly to AI-powered scientific discovery by lowering key barriers in automated literature data extraction and enriching structured knowledge databases.

METHODS

VisualHeist model fine-tuning

We downloaded 110 PDF articles and converted them into image files using the pdf2image library. Subsequently, bounding boxes around each figure-caption and table-heading pair were manually annotated using COCO Annotator,[53] with footnote descriptions also included within the bounding box. The annotated dataset was combined with the fine-tuning dataset from the TF-ID model,[33] resulting in a final dataset comprising 3,518 PDF pages, of which 3,424 included bounding box annotations, while 94 did not. Finetuning of Microsoft's Florence-2 model (Florence-2-large) was conducted over 12 epochs, with a learning rate of 5×10^{-6} and a batch size of 4.

Construction of evaluation datasets

To evaluate the performance of VisualHeist in figure and table segmentation, we manually curated a diverse evaluation dataset consisting of 121 literature articles covering a range of topics, including organic and inorganic chemistry, atmospheric science, batteries, materials science, MOFs, biology, and science education. These PDFs, published between 1949 and 2025, include both main articles and supplemental materials (refer to Table S1 for the complete DOI list).

We also additionally curated another collection of 98 literature articles reporting novel reaction methodologies that spans three distinct chemical domains: organic electrosynthesis, photocatalysis, and organic synthesis. We named this dataset as MERMaid-100 for future reference by other researchers. The full DOI list is available as Table S2 and can be downloaded from https://doi.org/10.5281/zenodo.14917751.[54] Using VisualHeist, figure-caption and table-heading pairs were first extracted as images, and a total of 104 pairs relevant to the task of reaction optimization were subsequently identified to assess the performance of DataRaider. MERMaid-100 was also used to benchmark the figure segmentation capability of OpenChemIE and PDFigCapX and the ground truth captions were compiled from the HTML versions of the articles using an automated Python script. All performance evaluations were performed manually by subject matter experts according to predefined criteria (detailed in each module's section; summarized in Table S12). While automated code for this step is not applicable, detailed evaluation results and raw outputs are included in the supplemental information and Zenodo repository to support reproducibility (refer to data and code availability).

VLM model selection

GPT-4o-2024-08-06 was selected as the VLM in DataRaider and KGWizard. We chose this model based on findings from our previous work,[15] where we benchmarked several leading VLMs for scientific visual tasks, namely reaction diagram parsing and reference label resolution. GPT-4V demonstrated competitive performance across these tasks and we therefore selected GTP-4o-0806, the latest publicly available multimodal model in the GPT-4 series at the time of investigation, for MERMaid. For details on comparative VLM performance, we refer readers to our prior work.[15]

Knowledge graph construction

For the construction of the knowledge graph, we used JanusGraph 1.1.0 to deploy the database system. This implementation was chosen due to its open license, maturity, and seamless Python integration through the gremlinpython library. Given the success of type-enforcement libraries such as Pydantic in improving the quality and security of LLM responses, we leveraged Python type annotations (i.e., the standard typing library) to define the data schema. These annotations establish an abstract layer specifying node types, the information they contain, and the relationships and constraints governing the edges (e.g., HasComment can only connect an Experiment node with a Comment node). By relying solely on the Python type system, these schemas provide a robust interface between the JSON data generated by the DataRaider module and the database API (gremlinpython). Additionally, data type validation is performed during node construction, ensuring that information adheres to the expected format before insertion into the database. The decision to use these schemas instead of Pydantic was made for transparency and simplicity; the raw code is publicly available in our repository, and we encourage users to analyze and adapt it according to their needs.

The translation from the DataRaider JSON output to the defined abstract type-based schema follows an iterative process involving GPT-4o-202408-06 as the LLM. The pre-training phase consists of the following steps:

(1) A minimal prompt with preset instructions is established as the basis (Note S7).
(2) The code defining the abstract representation of the graph is embedded within the prompt.
(3) A unique identifier for the JSON file, based on the DOI of the processed article, is generated and embedded into the prompt. This step ensures the uniqueness of articles and their corresponding experiments.
(4) The database is queried to retrieve unique node values, which are then dynamically incorporated into the prompt. This enforces consistency by guiding the model to select existing values while allowing the creation of new nodes, if necessary, thereby preventing data redundancy.

Once the pre-training phase is complete, experiments from the DataRaider JSON file are individually processed by the pre-trained model. The LLM then generates a serialized JSON representation of the predefined nodes and edges identified in each experiment. These outputs are subsequently deserialized into node and edge objects. During deserialization, a type-checking mechanism ensures that all values conform to their expected types. If a type mismatch is detected, an error is raised, halting the processing of subsequent experiments within the same article. Successfully validated nodes and edges are then integrated into the graph database.

This workflow is repeated for each DataRaider JSON file extracted from the bibliographic database, dynamically switching to the appropriate schema depending on the context. This ensures that the database evolves organically with newly processed data. However, the process is relatively slow due to the dataset's size and the LLM's response speed. To improve efficiency, articles are processed in parallel using a dynamic-size batch algorithm. Our algorithm starts with batches of 1 and slowly increases the batch size up 30.

This step is crucial to assure that, during the first batches, different values are not repeated in parallel runs. Although our batching method may introduce some redundant values, it represents a balanced trade-off between redundancy minimization and processing speed.

Once the graph is complete, the database is exported in GraphML format, facilitating data sharing and enabling visualization and analysis using specialized tools. In this work, we used the Gephi tool to generate the graph figures.

Structure-based LLM transformations

Let A be the category of structured texts and B the category of data types. Each object $X \in A$ is a structured text and each object $Y \in B$ is a data type. For any (X, Y) pair, we define a parser (morphism):

$$p_{X,Y} : X \rightarrow Y. \qquad \text{(Equation 1)}$$

We collect all such parsers into a single family:

$$P = \{ p_{X,Y} \mid X \in A, Y \in B \} \qquad \text{(Equation 2)}$$

Traditionally, each parser in P requires a specific implementation:

$$\forall (X, Y) \in A \times B,\ p_{X,Y} \in P \Rightarrow \exists f_{X,Y} : X \rightarrow Y. \qquad \text{(Equation 3)}$$

Here, $f_{X,Y}$ is the concrete logic transforming *values of structured text* X into values of type Y. However, implementing $p_{X,Y}$ can be both complex and time-consuming. To alleviate this burden, we employ an LLM that serves as a higher-order function:

$$g_{A,B} : A \times B \rightarrow (X \rightarrow Y). \qquad \text{(Equation 4)}$$

Given a structured text $X \in A$ and a data type $Y \in B$,

$$p_{X,Y} = g_{A,B}(X, Y) : X \rightarrow Y, \qquad \text{(Equation 5)}$$

they are automatically synthesized by the LLM using the definitions of X and Y, significantly simplifying the generation of each parser.

In our implementation, the type X is represented as:

$$X \sim (S, x),\ \text{where } S \text{ is a set of instructions and } x \in X. \qquad \text{(Equation 6)}$$

Meanwhile, Y is represented by Python code that defines its data type. In our framework, S is not merely a description of the data type X, but also a set of human-readable instructions detailing *how* to operate on values of X. Essentially, S can encode domain-specific knowledge, transformations, or parsing rules that guide the conversion process. Hence, when the LLM reads S, it gains insights into both the *structure* of X and the *actions necessary to process or transform* X's values before mapping them into the target type Y. This additional procedural information in S helps ensure that the automatically synthesized parser $p_{X,Y}$ can accurately handle X in all relevant use cases. KGWizard implements this behavior for a fixed X (namely, JSON from DataRaider) and a collection of Y specifications that define the structure of graph databases for different domains. Although our current implementation targets graph databases, the approach is not inherently restricted to them and can be adapted to other database paradigms, e.g., relational databases.

Hardware and computational resources

VisualHeist fine-tuning and inferences were performed using an Intel Xeon CPU w72495X (48 cores @ 4.60 GHz), and two NVLINK Nvidia RTX A6000 GPUs (48GB VRAM). Each inference requires 4.5 s on average. Performance evaluation of DataRaider was performed using a 13th Gen Intel(R) Core(TM) i7-1360P CPU (12 cores @ 2.20 GHz). Each inference requires 41.3 s on average. The total elapsed time taken to process a curated JSON file with KGWizard is on average 110.6 s. In terms of cost, the average processing cost per image for DataRaider and KGWizard is $0.051 and $0.40, respectively.

RESOURCE AVAILABILITY

Lead contact

Further information should be directed to and will be fulfilled by the lead contact, Alán Aspuru-Guzik (alan@aspuru.com).

Materials availability

This study did not generate any new materials.

Data and code availability

- The source code of our automated workflow extraction code, together with additional data mining scripts, are available at our GitHub repository: https://github.com/aspuru-guzik-group/MERMaid.
- A frozen version of the repository is deposited at Zenodo: https://doi.org/10.5281/zenodo.15025415.[55]
- The prompts used in this work, together with the raw responses from the tested VLM models are uploaded to Zenodo: https://doi.org/10.5281/zenodo.14917751.[54]
- Any additional information required to reanalyze the data reported in this paper is available from Shi Xuan Leong (shixuan.leong@utoronto.ca) upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, S.X.L., S.P.-G., and A. A.-G.; data curation, S.X.L.; formal analysis, S.X.L., S.P.-G., and B.W.; investigation, S.X.L.; methodology, S.X.L. and S.P.-G.; validation, S.X.L.; visualization, S.X.L.; software, S.X.L., S.P.-G.,

and B.W.; writing – original draft, S.X.L.; writing – review & editing, S.X.L., S. P.-G., and A.A.-G.; supervision, S.P.-G. and A.A.-G.; funding acquisition, A. A.-G.; resources, A.A.-G.; project administration, A.A.-G.

DECLARATION OF INTERESTS

A.A.-G is an employee of NVIDIA.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used ChatGPT to improve language and grammar. After using this tool or service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

SUPPLEMENTAL INFORMATION

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